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Hypervariable 3′ UTR region of plant LTR-retrotransposons as a source of novel satellite repeats

Jiří Macas *, Andrea Koblížková, Alice Navrátilová, Pavel Neumann

Biology Centre ASCR, Institute of Plant Molecular Biology, Branišovská 31, České Budějovice, CZ-37005, Czech Republic

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ABSTRACT

The repetitive sequence PisTR-A has an unusual organization in the pea (*Pisum sativum*) genome, being present both as short dispersed repeats as well as long arrays of tandemly arranged satellite DNA. Cloning, sequencing and FISH analysis of both PisTR-A variants revealed that the former occurs in the genome embedded within the sequence of Ty3/gypsy-like Ogre elements, whereas the latter forms homogenized arrays of satellite repeats at several genomic loci. The Ogre elements carry the PisTR-A sequences in their 3′ untranslated region (UTR) separating the *gag-pol* region from the 3′ LTR. This region was found to be highly variable among pea Ogre elements, and includes a number of other tandem repeats along with or instead of PisTR-A. Bioinformatic analysis of LTR-retrotransposons mined from available plant genomic sequence data revealed that the frequent occurrence of variable tandem repeats within 3′ UTRs is a typical feature of the Tat lineage of plant retrotransposons. Comparison of these repeats to known plant satellite sequences uncovered two other instances of satellites with sequence similarity to a Tat-like retrotransposon 3′ UTR regions. These observations suggest that some retrotransposons may significantly contribute to satellite DNA evolution by generating a library of short repeat arrays that can subsequently be dispersed through the genome and eventually further amplified and homogenized into novel satellite repeats.

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1. Introduction

Repetitive DNA sequences are classified based on the genomic organization of their units as either dispersed or tandem repeats. In higher plants, both dispersed and tandem repeats are abundant and differential amplification of their various families contribute significantly to the extraordinary genome size variation found between plant taxa (Flavell et al., 1974; Neumann et al., 2006; Zuccolo et al., 2007). The dispersed repeats are represented by various classes of mobile elements, which have the ability to transpose their copies within the genome either by copy-and-paste (retrotransposons) or cut-and-paste (DNA transposons) mechanisms. They typically encode the proteins facilitating their own transposition and employ various strategies to maintain or increase their copy numbers in the genome (Feschotte et al., 2002; Kazazian, 2004; Wicker et al., 2007). Tandem repeats, however, usually have no coding capacity and the mechanisms of their evolution are much less well understood. Replication slippage and recombination probably play a major role in shaping the arrays of tandem repeats with short monomers, the microsatellites (monomers of 1–5 bp) and minisatellites (monomers up to tens of nucleotides) (Schlotterer, 2000; Richard et al., 2008). Evolution of the satellites, which are composed of monomers tens to thousands nucleotides long, is supposed to be driven by a concurrent action of various processes including unequal crossing over, gene conversion and eccDNA formation (Stephan, 1986; Elder and Turner, 1995; Navrátilová et al., 2008). Unlike micro- and minisatellites, satellite DNA is often highly amplified in plant genomes and occurs in long uninterrupted arrays up to several megabases long (Cheng et al., 2002; Tek and Jiang, 2004).

Due to the principally different amplification mechanisms of transposons and satellite repeats, genomic evolution of these two repeat classes has mostly been investigated independently. However, there is growing evidence that mobile elements can be involved at various stages of tandem repeat evolution. For example, a considerable part of micro- and minisatellites present in eukaryotic genomes are embedded within mobile elements which may facilitate their amplification and dispersion in the genome (Inukai, 2004; Smýkal et al., 2009). There are also reports describing satellite repeats originating from fragments of retroelement sequences (Langdon et al., 2000; Cheng and Murata, 2003; Tek et al., 2005). They correspond to retroelement regions which originally did not have tandem structure, like the *gag* gene in the case of the 250 bp satellite in *Aegilops speltoides* (Cheng and Murata, 2003), or include large parts of retroelements combined with non-retroelement sequences to form satellites with exceptionally long monomers (Langdon et al., 2000; Tek et al., 2005). Therefore, it is likely that retrotransposon sequences

Abbreviations: FISH, fluorescence in situ hybridization; LTR, long terminal repeat; ORF, open reading frame; PBS, primer binding site; RT, reverse transcriptase; UTR, untranslated region.

* Corresponding author. Tel.: +420 38 7775516; fax: +420 38 5310356.
E-mail address: macas@umbr.cas.cz (J. Macas).

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played only a passive role during the origin of these satellites, being subjected to random recombination and amplification.

In this work, we describe a novel pathway of satellite DNA formation from mobile elements, which is based on amplification and homogenization of tandem repeats frequently generated within the hypervariable 3′ UTR region of LTR-retrotransposons. The evidence for this mechanism was revealed during characterization of PisTR-A repeats present in the pea (*Pisum sativum*) genome. These repeats were identified during our previous studies of pea repetitive DNA (Neumann et al., 2001), and were classified as satellite DNA based on a ladder-like patterns detected on Southern blots of digested genomic DNA. However, FISH experiments with PisTR-A probe produced dispersed hybridization signals on mitotic chromosomes (Neumann et al., 2001), contrary to the limited number of strongly hybridizing loci that would be expected for clusters of satellite DNA. The new data presented here demonstrate that this observation was due to the presence of short stretches of PisTR-A sequences within 3′ UTR of Ogre retrotransposons dispersed in the pea genome, and that these short PisTR-A repeats were amplified into long satellite arrays at several genomic loci. It was also found that the formation of tandem repeats in 3′ UTRs is a feature typical of the Tat lineage of Ty3/gypsy elements, which, in addition to Ogre elements, includes a number of other retrotransposons abundant in various plant species.

2. Materials and methods

2.1. Plant material and cloning procedures

Seeds of garden pea (*P. sativum* L.) were obtained from the Plant Breeding Station at Boršov, Czech Republic. Total genomic DNA was extracted from leaves as described by Dellaporta et al. (1983). Shotgun libraries of pea genomic DNA cloned to plasmid and cosmid vectors (Neumann et al., 2003) were screened with a 211 bp PisTR-A fragment described by Neumann et al. (2001). The screening of all libraries was done by colony hybridization, using AlkPhos Direct kit (Amersham) for probe labeling and signal detection (hybridization and washing temperatures were 55 °C). Positive clones were subjected to sequencing either directly (the plasmid library) or following random or directional subcloning of smaller fragments. Sequencing reads were processed and assembled using Staden Package software (Staden, 1996). Size estimation of PisTR-A array within c711 was done by digesting the clone with restriction enzymes which do not cut in PisTR-A sequences (AluI, EcoRI, SalI) and analyzing resulting fragments using filed inversion gel electrophoresis (FigeMapper, BioRad). Electrophoresis conditions were set to resolve DNA fragments in 5–100 kb range, and the fragment length was estimated by comparison to 5 kb size standard (BioRad).

2.2. Fluorescence in situ hybridization (FISH)

In situ hybridization experiments were performed on squash chromosome preparations (Leitch et al., 1994) prepared from synchronized pea root tips as described (Neumann et al., 2002). The satellite variant of PisTR-A was detected by a LNA-oligonucleotide probe PisTR-A_SAT1_LNA (5′-ctc ttc aGg aTg aTg tCg gtc g-3′, LNA bases are in uppercase letters) which was biotin-labeled at its 5′ end (Sigma-Proligo). The probe PisTR-A/2, detecting both satellite and dispersed PisTR-A repeats, was prepared by PCR amplification of the cloned sequence (AF300839) and labeled with biotin-16-dUTP (Boehringer Mannheim) during the reaction. A 2.5 kb subclone (Ogre-PA-orf1-gag) of the cosmid c700 corresponding to adjacent parts of ORF1 and *gag* coding regions of Ogre-PA was labeled with biotin using nick translation (Leitch et al., 1994) and used to detect Ogre elements. Chromosome denaturation was carried out in a 1× PCR buffer (Promega) supplemented with 4 mM $MgCl_2$ for 2 min at 94 °C. The hybridization mix consisted of 2× SSC, 100 ng/μl sheared calf thymus DNA, 0.125% SDS, 50% formamide and 0.5 ng/μl PisTR-A/2 and Ogre-PA-orf1-gag probe or 0.1 μM PisTR-A_SAT1_LNA. Before hybridization, the mix was denatured at 76 °C for 15 min. Hybridization was performed overnight at 37 °C, and post-hybridization washes were performed in 2× SSC, 50% formamide at 42 °C for 15 min. The probes were detected with streptavidin-Alexa Fluor 568 (Invitrogen). Chromosomes were counterstained with DAPI (4′, 6-diamidino-2-phenylindole) and examined using a Nikon Eclipse 600 microscope. The images were captured using a CCD camera and analyzed using Lucia software (Laboratory Imaging).

2.3. Sequence analysis

Potential LTR-retrotransposons were identified in a set of plant genomic sequences (GBPLN set, 3.8 Gbp, downloaded from GenBank in March 2009) using LTR-finder (Xu and Wang, 2007). The program was run with the parameters –L 7000 –r 12 –w 2 –F 11110100000, using *Arabidopsis* tRNA database (Lowe and Eddy, 1997) to identify PBS regions. The internal element regions were extracted based on the program output and searched using Tandem Repeats Finder (Benson, 1999) in order to identify tandem repeats with monomers of at least 30 bp occurring as arrays at least 150 bp long. The elements containing tandem repeats were grouped based on the similarity of their repeats in order to facilitate their analysis. The grouping was performed with TGICL (Pertea et al., 2003) using minimal sequence similarity of 94% over at least 50 nucleotides as the clustering parameters. Sequence manipulations including output parsing from the programs listed above and various subsequent analyses were performed using custom made scripts written in BioPerl (http://www.bioperl.org). The individual groups of sequences were examined manually in order to verify that they represent correctly identified LTR-retrotransposons and their mutual similarities were investigated using standalone BLAST (Altschul et al., 1997) and Dotter (Sonnhammer and Durbin, 1995) programs. Phylogenetic analysis of identified retrotransposons was done based on multiple sequence alignments of reverse transcriptase domains created using the ClustalX program (Thompson et al., 1997). The phylogenetic tree was built using neighbor-joining algorithm implemented in ClustalX and drawn using the iTOL online tool (Letunic and Bork, 2007).

3. Results

3.1. Tandem repeat PisTR-A is a part of the 3′ UTR of an Ogre retrotransposon and it is also highly amplified at several genomic loci

To investigate sequence variability of PisTR-A repeats in the pea genome, a short-insert shotgun genomic library was screened using the previously isolated PisTR-A clone (Neumann et al., 2001) as a probe. Sequencing 16 randomly picked clones revealed that only one was composed entirely of tandemly arranged PisTR-A repeats, whereas the rest included only short or imperfect PisTR-A repeats. Sequences surrounding the imperfect PisTR-A repeats were similar among these clones, suggesting they originated from the same repetitive element. In order to retrieve a full-length copy of this element, a pea cosmid library was screened and six clones were partially or completely sequenced. There was no intact element identified in these clones, due to their trimming by cloning or interruptions by nested insertions of other elements. However, analysis of the acquired sequence data allowed us to reconstruct the full-length repeat and classify it as a Ty3/gypsy-like Ogre element (Macas and Neumann, 2007) designated Ogre-PA.

The structure of the reconstructed element is depicted on Fig. 1A, including the position of the PisTR-A sequence which was found to be located within the element's 3′ untranslated region (3′ UTR) separating the *pol* region from the 3′ LTR. Comparative analysis of this region in elements from individually sequenced clones revealed

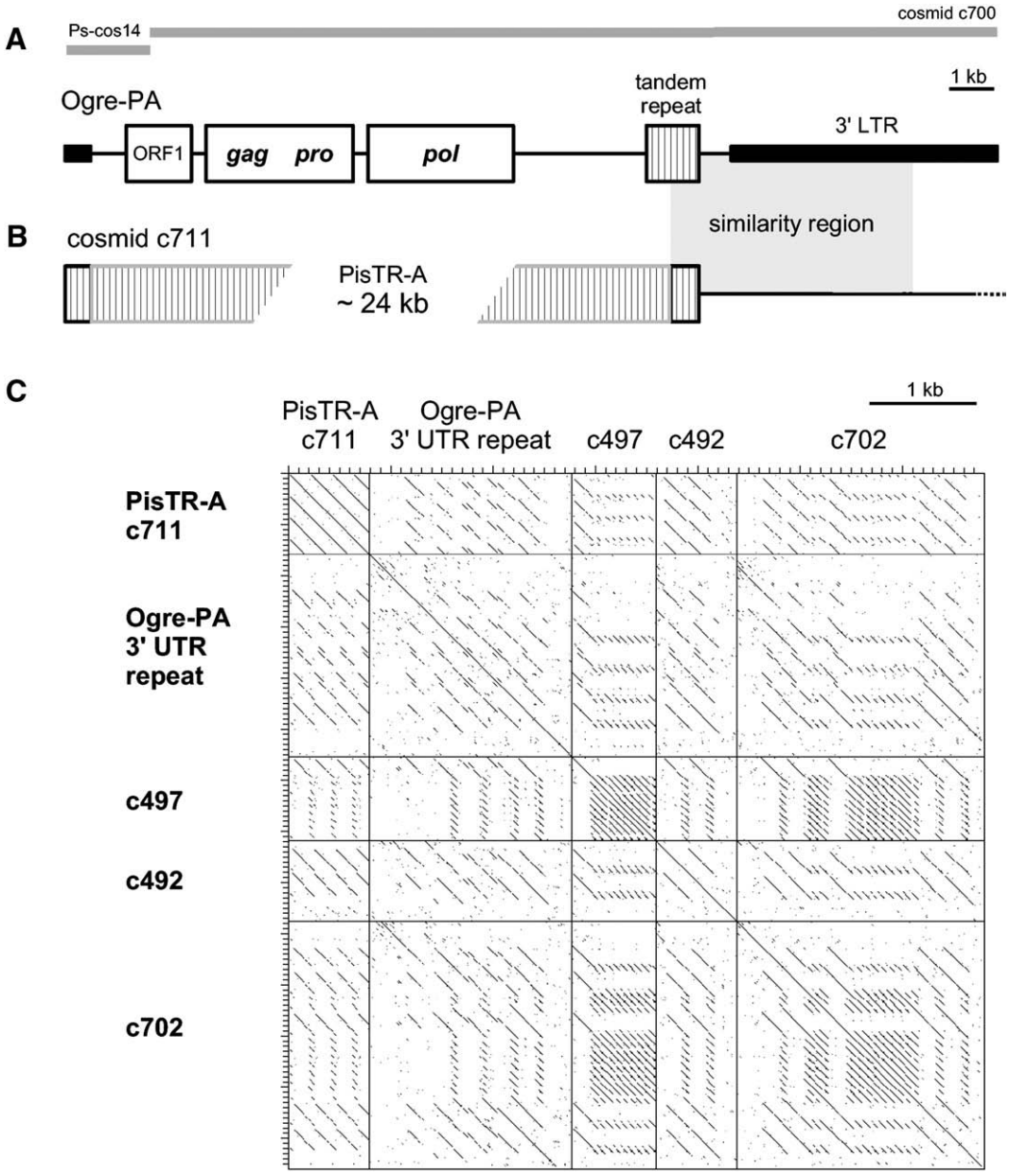


Fig. 1. Position and variability of tandem repeats in pea Ogre sequences. (A) A schematic representation of an Ogre-PA element containing tandem repeats with similarity to PisTR-A. An almost complete element lacking only a part of its 5′ LTR was reconstructed from sequences in clones c700 (accession no. GQ267464) and Ps-cos14 (AY299397, positions 18845–17013) (gray lines above the scheme). The LTRs and ORFs are displayed as black and open boxes, respectively. The tandem repeat array is shown as a hatched box. (B) Insert structure of the cosmid clone c711 (GQ267465) which is mostly made up of a continuous PisTR-A array adjacent to sequences similar to Ogre 3′ UTR and LTR sequences (similarity region is shaded). (C) Dot-plot similarity comparison of both the satellite variant of PisTR-A and the PisTR-A-related, 3′ UTR localized tandem repeat from Ogre-PA to tandemly repeated regions of selected clones (accession nos.: c497, FI903121; c492, FI903114; c702, GQ267466).

that it was highly variable and included tandem repeats originating from various parts of the 3′ UTR sequence and differing in their monomer lengths (Fig. 1C). In addition, a cosmid clone (c711) was identified which included a long array of PisTR-A repeats adjacent to a 9.6 kb sequence with partial similarity to the 3′ UTR and LTR sequences of Ogre-PA (Fig. 1B). Although it was not possible to assemble the complete insert sequence due to high sequence similarity of individual PisTR-A repeats (91–100%), restriction analysis combined with pulsed field electrophoresis revealed that the PisTR-A sequences form a contiguous array spanning about 24 kb (data not shown). As this region was 5′ truncated by cloning, it can be assumed that the corresponding genomic locus contained an even longer array of PisTR-A, which due to its length and high monomer homogenization could be classified as a satellite repeat.

In order to verify the presence of satellite arrays of PisTR-A sequences in the pea genome, we attempted to detect these repeats using FISH on mitotic chromosomes. A probe specific for the satellite sequence variant was designed based on comparison of PisTR-A repeats present in the cosmid c711 to those from other clones containing elements with only short stretches of the repeat. This probe produced hybridization signals typical for satellite DNA, appearing as strong dots localized at limited number of genomic loci, and a number of additional significantly weaker dispersed signals which were probably caused by partial cross-hybridization to other PisTR-A repeats (Fig. 2A–C). On the contrary, the probe with similarity to both types of PisTR-A repeats produced hybridization signal dispersed on all chromosomes (Fig. 2D–F). This signal was similar to that obtained with a probe derived from parts of the Ogre coding

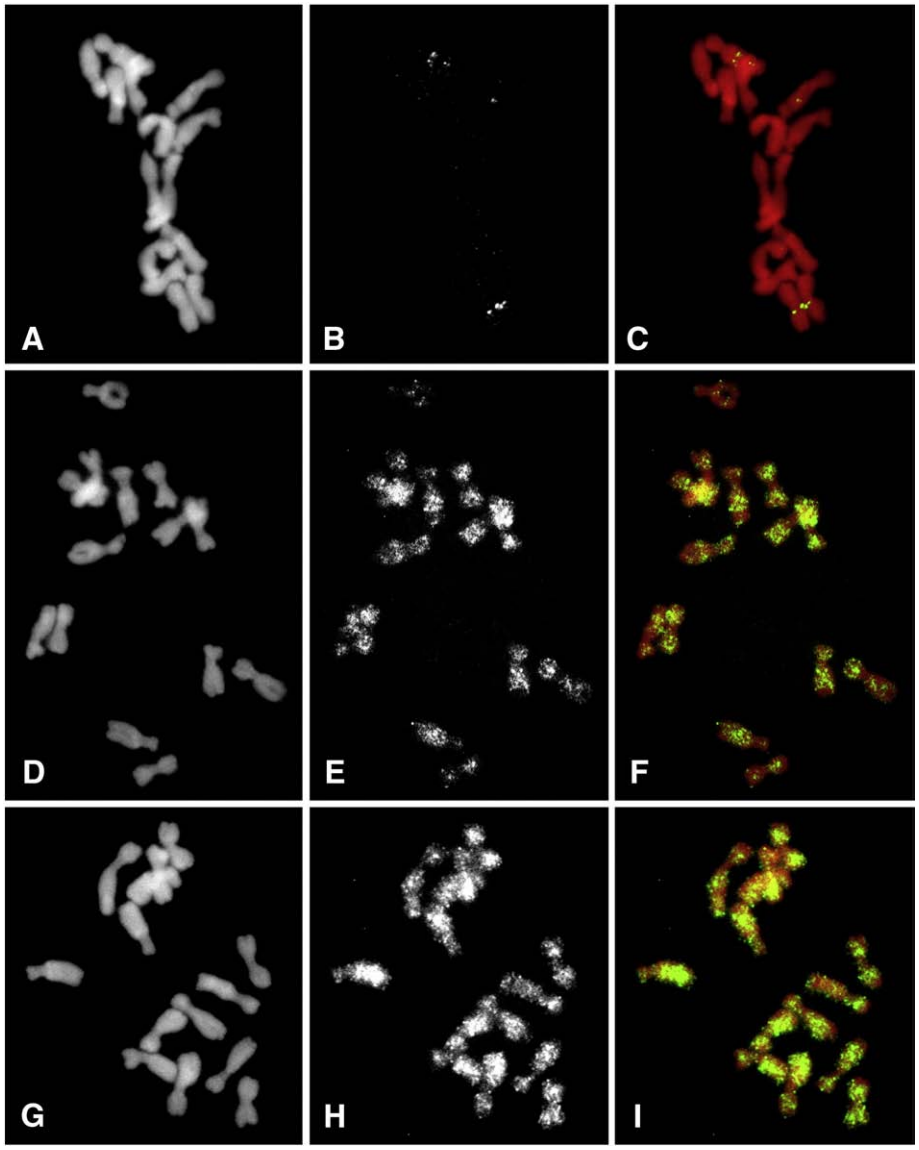


Fig. 2. Localization of PisTR-A sequences on *P. sativum* chromosomes ($2n = 14$) using FISH. The metaphase chromosome preparations were hybridized to the oligonucleotide probe PisTR-A_SAT1_LNA specific for the satellite variant of the repeat (A–C) or to the probe detecting all PisTR-A repeats (D–F). The probe derived from parts of Ogre ORF1-gag coding regions was used as a control to localize Ogre sequences (G–I). The panels A, D, G show chromosomes counterstained with DAPI, while the probe signals are shown on B, E, H. The panels C, F, I then show merged DAPI (red) and probe (green) channels.

regions (Fig. 2G–I) and thus was probably generated mainly by PisTR-A sequences present within Ogre elements.

3.2. A hypervariable 3′ UTR including tandem repeats is a typical feature of the Tat lineage of plant Ty3/gypsy retrotransposons

To investigate whether the occurrence of tandem repeats in 3′ UTR regions is common in plant LTR-retrotransposons, we performed their bioinformatic detection in the plant genomic sequences. A total of 3.8 Gbp of plant sequence data available from GenBank was first analyzed by LTR-finder (Xu and Wang, 2007) to identify LTR-retrotransposons. Sequences of the 25,128 identified potential elements were then screened for the presence of tandem repeats with monomers of at least 30 bp and spanning at least 150 bp. Only the internal element regions (excluding LTRs) were subjected to this screening as our search was primarily focused to identification of tandem repeats within 3′ UTR regions. A total of 5945 elements contained tandem repeats in their internal sequence regions and preliminary analysis based on the elements with detectable reverse transcriptase domains revealed that most of these belonged to the Tat lineage of Ty3/gypsy elements (Llorens et al., 2008). The percentage of elements with tandem repeats was 41.2% in this lineage, while it was only 9.5% in other lineages of gypsy elements, and 7.5% in Ty1/copia elements. The elements were grouped based on sequence similarities of their tandem repeats and further examined to determine the exact position and composition of tandem repeats in their sequences and to verify the element type based on their RT and PBS sequences. This investigation revealed that the tandem repeats present in 3′ UTRs occurred solely in the Tat elements. In addition to the pea Ogre elements, they included three families of previously described elements known to contain tandem repeats — Grande and Cinful from *Zea mays* (Martínez-Izquierdo et al., 1997; Sanz-Alferez et al., 2003) and Retand from *Silene latifolia* (Kejnovský et al., 2006). Further, a number of new elements were identified in several plant genomes including *Medicago truncatula*, *Populus trichocarpa*, *Vitis vinifera* and *Oryza sativa* (Table 1). Phylogenetic analysis of these sequences revealed their close relationship to various elements within the Tat lineage (Fig. 3). It should be noted that there were also many non-autonomous elements that could not be subjected to phylogenetic analysis due to the lack of RT domains (Table 1). However, most

Table 1
A list of selected elements typical for individual groups of identified LTR-retrotransposons.

Element[a]	Length [bp]	TSD[b]	RT[b]	Tandem repeat		GenBank acc. no.	Element position	Family[d]
				Group	Size[c]			
At1	6264	+	−	61	2.2×82	AC006250	6496–233	
Gm1	16915	+	−	STR	20.5×123	FJ197996	16984–1	
Gm2	18974	+	+	STR	(13.7×110)	EF623859	54292–73265	
Gm3	16546	+	−	STR	(7.8×100)	AC170861	107996–124541	
Gm4	19538	+	+	STR	(7.4×100)	EU450800	74393–93930	
Hv1	13675	+	+	H_70	(12×70)	AY642926	139519–125845	
Lj1	3765	+	−	17	3.3×65	AP004466	90854–94618	
Lj2	4752	+	−	139	4.0×73	AP010286	41853–37102	
Mt1	15665	+	+	4	3.1×174	EU306659	80275–95939	
Mt2	8065	+	−	33	8.7×103	CT961059	17956–26020	
Mt3	12780	+	+	37	5.0×250	CU914132	65103–52324	
Mt4	17426	+	−	37	6.8×32	CU633191	24982–7557	
Mt5	16203	+	+	42	3.3×106	AC137824	59807–76009	
Mt6	11512	+	−	57	3.9×83	CU914761	77178–65667	
Mt7	5597	+	−	71	9.1×121	CT868735	45120–50716	
Mt8	11103	+	−	75	5.9×96	AC149306	52924–41822	
Mt9	8109	+	−	109	1.8×192	CU463043	1843–9951	
Mt10	16724	+	+	113	5.8×95	AC130653	67907–84630	
Mt11	11528	+	−	121	3.1×160	AC203557	84634–96161	
Mt12	16637	+	+	128	11.2×53	AC129090	107013–123649	
Os1	12852	+	+	1	1.8×192	AB097479	4632–17483	Osr27
Os2	4729	+	−	6	9.9×71	EU155083	6691–11419	Bajie
Os3	4371	+	−	7	5.0×78	AP008218	4794145–4789775	
Os4	13857	+	+	8	2.5×69	AC104616	85032–71176	Osr30
Os5	5644	−	−	35	3.0×147	AC084884	23461–29104	
Os6	11369	−	−	35	2.3×73	AP004883	9460–20828	
Os7	4737	+	−	35	3.0×145	AP008212	15424235–15419499	
Os8	3984	−	−	35	1.8×133	AP008217	25340359–25344342	Osr29
Os9	13675	+	+	59	17.8×53	AP003376	71487–57813	
Os10	13845	+	+	64	4.3×148	AP008218	12554500–12568344	
Os11	12962	+	+	76	3.8×151	AL606587	125250–138211	
Os12	4314	+	−	81	2.5×372	CR855036	5315–1002	
Os13	8959	+	+	86	4.4×43	AC135459	48341–57299	Osr29
Os14	5816	+	−	97	3.6×344	AL731786	16896–22711	
Os15	3833	+	−	101	8.8×92	AP008216	5179637–5183469	
Os16	12166	−	−	105	2.4×310	AP008213	14065037–14052872	
Os17	4293	+	−	106	4.1×78	AL713945	75919–80211	
Os18	22886	−	−	132	2.8×146	AC092778	40837–17952	
Os19	8870	+	−	134	8.8×78	AL606444	8227–17096	
Os20	16072	+	+	136	3.6×42	AP002867	53614–37543	Osr27
Os21	8645	+	−	2	31.5×89	AB023482	92454–83810	Dasheng
Ps1	21467	−	+	98	2.9×68	AY299398	35967–14501	Ogre-PS1
Pt1	4046	+	−	15	3.0×71	AC149296	50824–46779	Ogre-PT1
Pt2	11806	+	−	26	8.2×101	AC182707	49197–61002	
Pt3	9675	+	−	26	11.8×63	AC209191	54620–64294	
Pt4	17726	+	+	30	2.9×444	AC187862	93221–75496	
Pt5	17076	+	+	30	(7.4×110)	AC149300	92345–109420	Ogre-PT3
Pt6	14194	+	+	44	5.7×89	AC213014	15229–1036	
Pt7	10138	+	−	49	15.1×38	AC184075	104475–94338	
Pt8	10276	+	−	49	2.7×114	AC212918	4532–14807	
Pt9	19716	+	+	68	13.5×199	AC216768	17600–37315	
Pt10	1556	+	−	135	6.0×84	AC183424	10046–8491	
Pt11	15923	+	−	147	31.1×30	AC209042	30264–14342	
Pt12	17799	+	+	0	11.5×61	AC149296	56339–74137	
Sb1	2761	+	−	123	2.8×80	AC196829	27437–24677	
Sil1	11073	+	+	0	(irregular)	DQ023669	123–11195	Retand-2
Sil2	3679	+	−	0	3.9×201	DQ023670	1434–5112	Retand-1
Sol1	22179	+	+	96	8.0×237	AC210359	10128–32306	
Ta1	12503	+	+	H_70	(7.7×73)	AM932684	139439–126937	
Tm1	12555	+	+	H_70	1.9×354	AY485644	236669–249223	
Vv1	17858	−	+	63	4.8×64	AM447791	20232–2375	
Vv2	8328	+	−	67	2.9×141	AM476031	2491–10818	
Vv3	16424	−	+	102	3.3×191	AM467778	6087–22510	
Zm1	9555	+	+	112	4.6×80	AC160211	61269–51715	
Zm2	7459	+	−	112	3.0×60	AY325816	68435–60977	
Zm3	12434	+	+	145	6.0×89	AF448416	82278–94711	

[a] Element names are derived from abbreviated species names: At, *Arabidopsis thaliana*; Gm, *Glycine max*; Hv, *Hordeum vulgare*; Lj, *Lotus japonicus*; Mt, *Medicago truncatula*; Os, *Oryza sativa*; Ps, *Pisum sativum*; Pt, *Populus trichocarpa*; Sb, *Sorghum bicolor*; Sil, *Silene latifolia*; Ta, *Triticum aestivum*; Tm, *Triticum monococcum*; Vv, *Vitis vinifera*; Zm, *Zea mays*.

[b] The presence of target site duplication (TSD) surrounding the element insertion and reverse transcriptase coding domain (RT) within its sequence is indicated by '+'.

[c] The size of the tandem repeat detected within an element is given as a number of repetitions × monomer size (in bp). These values were determined by the TRF program except for those given in parentheses which were determined manually. In case of the simultaneous occurrence of several repeat arrays within a single element, only the one giving the best TRF score is listed.

[d] Indicates identity or strong similarity to known elements or their families.

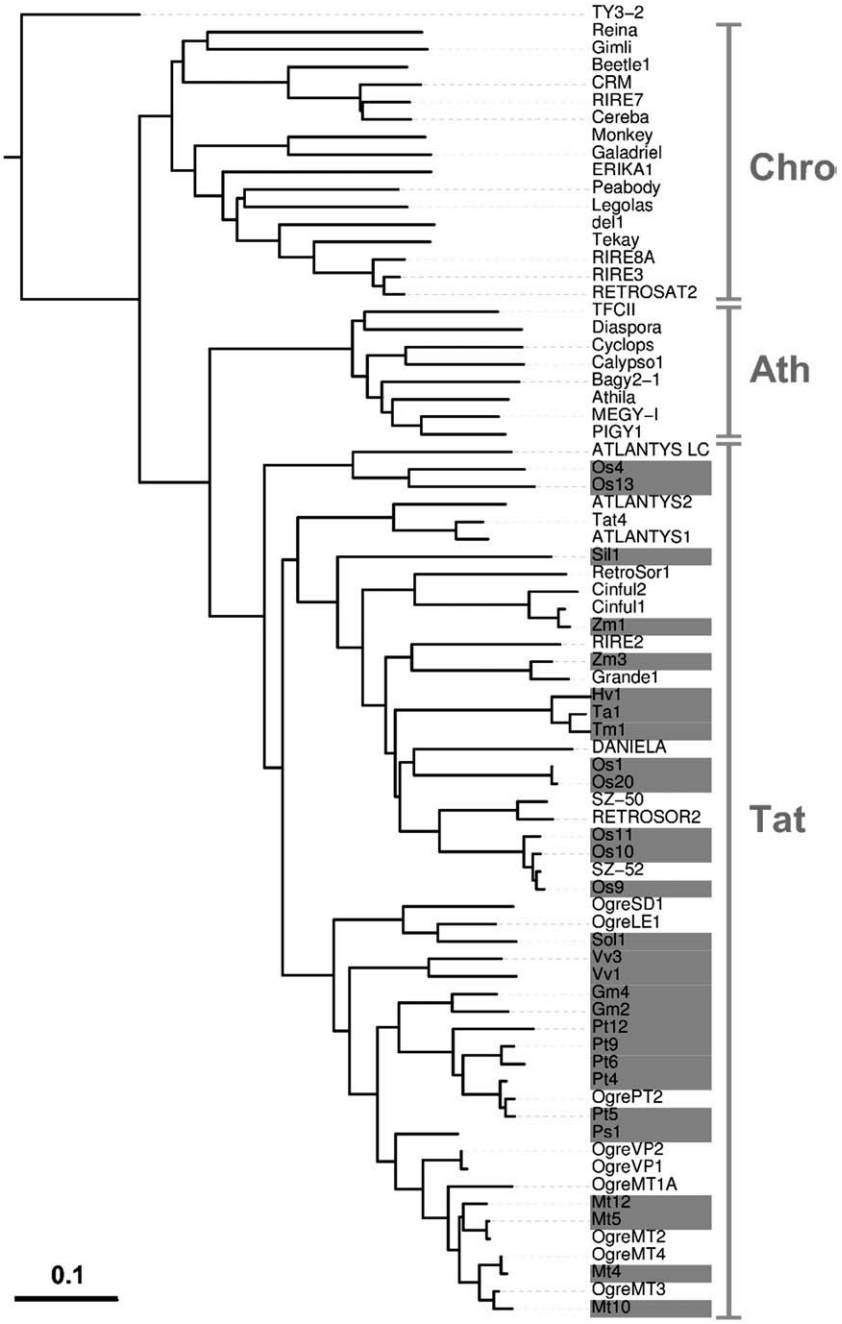


Fig. 3. Phylogenetic analysis of Ty3/gypsy LTR-retrotransposons based on multiple alignment of their reverse transcriptase domains. The elements identified in this study are shaded and their names correspond to those used in Table 1. The tree was rooted using retrotransposon Ty3-2 from *Saccharomyces cerevisiae*. Chromovirus (Chro), Athila (Ath) and Tat lineages of the plant Ty3/gypsy retrotransposons were distinguished according to Llorens et al. (2008). The scale bar indicates an evolutionary distance of 0.1 amino acid substitutions per position.

of them could also be assigned to the Tat lineage based on their similarities to the autonomous elements or due to the presence of PBS sequences complementary to tRNA-Arg, which is a feature typical for this group.

Comparative analysis of the identified 3′ UTR tandem repeats and the elements carrying them revealed that, similarly to the Ogre elements, this region exhibits high variability with respect to the sequence and monomer size of the tandem repeats. The repeat monomers varied from 30 to 444 bp (Table 1) and some elements contained up to three different tandem repeats. The length of the repeat arrays also varied, from several hundreds of nucleotides to over 10 kb. The tandem repeats sequences were species-specific and often confined to a specific element subfamily within the species. Interestingly, there were also several cases where certain tandem repeats appeared to be conserved within otherwise unrelated 3′ UTR sequences. These observations are demonstrated on Fig. 4, showing structural and sequence comparison of six elements identified in the genome of black cottonwood (*P. trichocarpa*) as an example.

3.3. Satellite repeats derived from the 3′ UTR sequences occur in several plant species

The high frequency and diversity of 3′ UTR tandem repeats found in elements from various species raised a question whether there are more satellites derived from these repeats in a way similar to PisTR-A in pea. Thus, we performed a sequence similarity search between all

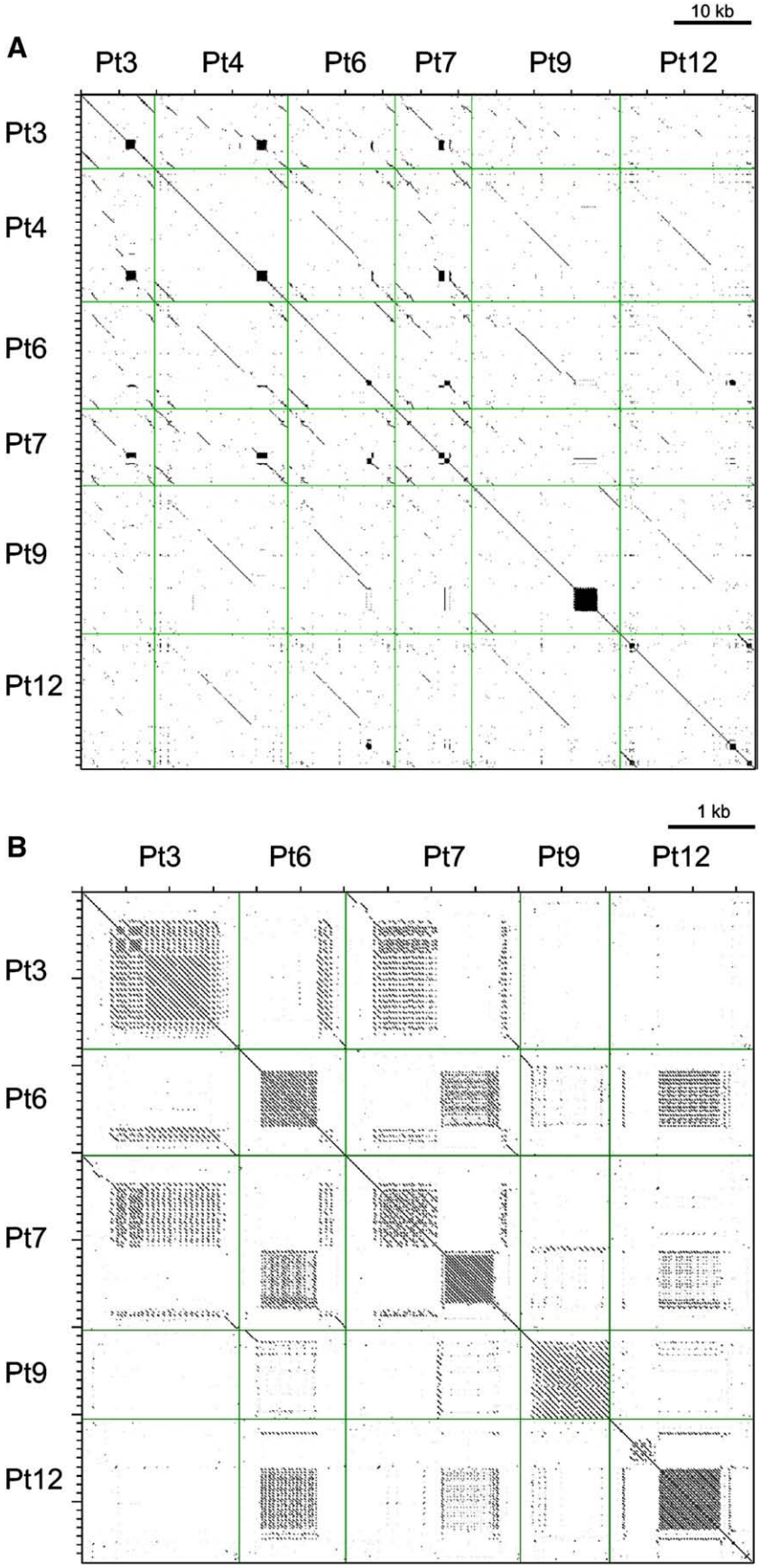


Fig. 4. Dot-plot comparison of selected retrotransposon sequences identified in the *P. trichocarpa* genome. (A) Comparison of complete element sequences, with repeat arrays appearing as black squares. A detailed view of these 3′ UTR repeat regions is provided in (B). The element names correspond to Table 1. The repeats from the element Pt4 are not shown in B as they are almost identical to Pt3. Only a part of the 5 kb long repeat array from Pt9 is magnified in panel B.

identified 3′ UTR repeats and a set of known plant satellites available from PlanSat database (Macas et al., 2002). Significant hits were found for the satellite repeat family STR-120 from soybean (Morgante et al., 1997) and for the pHvMWG2314 satellite sequence from barley (Busch et al., 1995).

The soybean satellite STR-120 is a repeat with monomers of 120 bp which was extensively characterized using molecular and cytogenetic approaches. STR-120 was found to be localized at two to eight chromosomal loci (based on the FISH stringency) and its clones were used to selectively label specific subsets of soybean chromosomes (Shi et al., 1996; Morgante et al., 1997). There is also a sequence variant of this repeat with a shorter monomer length, STR-102 (Lin et al., 2005), which was also found to produce FISH patterns characteristic of satellite DNA and was estimated to form clusters up to 435 kb long in the soybean genome (Lin et al., 2005). Here we have identified a family of soybean retrotransposons, including the elements Gm1 and Gm2 listed in Table 1, which contain tandem arrays of STR120/102 repeats in their 3′ UTRs. The STR repeats were not conserved in this retrotransposon family as there were also other elements, Gm3 and Gm4, where STR sequences were replaced by other tandem repeats, although these elements shared sequence similarity of their LTRs as well as parts of internal regions with Gm1 and Gm2 (see Fig. 3 for positions of Gm2 and Gm4 within the phylogenetic tree).

The repeat pHvMWG2314 was first described in barley as a 73 bp-monomer satellite with partially dispersed genomic organization (Busch et al., 1995). A subsequent study, including its chromosomal localization using FISH, revealed its amplification in various Poaceae species and accumulation at distinct sites of several wheat chromosomes (Busch et al., 1996). The LTR-retrotransposons carrying 3′ UTR tandem repeats with similarity to pHvMWG2314 were identified in barley (Hv1), and interestingly also in *Triticum monococcum* and *T. aestivum* (Tm1 and Ta1, Table 1). This is the only case in our study where the same tandem repeat was found in elements from different species.

4. Discussion

Satellite DNA represents one of the most dynamic components of plant genomes, undergoing rapid changes in abundance and sequence composition (Tek et al., 2005; Macas et al., 2006). While there is evidence for several molecular mechanisms which could facilitate sequence amplification and homogenization of satellite DNA (Elder and Turner, 1995; Dover, 2002), there is considerable uncertainty about the origin of short arrays of tandem repeats, which presumably represent the first step in satellite DNA evolution. Theoretical models and computer simulations suggested that these repeats could be generated by unequal crossing over of random sequences (Smith, 1976). It was also proposed that short tandem repeats could originate due to sequence directed mutagenesis (Fieldhouse and Golding, 1991). However, there is an increasing number of reports demonstrating that a substantial proportion of tandem repeats in eukaryotic genomes occur within transposable elements (Hisatomi et al., 1997; Inukai, 2004; Smýkal et al., 2009). Although a part of tandem repeats present in transposons could originate by chance and thus occur at various positions, here we show that Tat-like retrotransposons are characterized by a specific hypervariable region of their sequences, the 3′ UTR, which frequently contains tandem repeats. Such frequent generation of diverse tandem sequences within a retroelement can have important consequences for the origin of satellite repeats, as it provides a library of short repeat arrays which can then be dispersed to various genomic loci due to replication and transposition of the element itself. These repeats may then be subsequently expanded into satellite DNA at genomic loci where appropriate molecular mechanisms are active. The data presented here has shown that such satellites exist in at least three plant species. However, this phenomenon may be much more common due to the fact that Tat elements are widespread in plant genomes and represent a major group of repeats in many species (Neumann et al., 2006; Macas et al., 2007; Zuccolo et al., 2007). This presumption is also supported by reports of additional repeats with both tandem and dispersed genomic organizations similar to PisTR-A from diverse taxa including the cycas *Zamia paucijuga* (Cafasso et al., 2003) and the legume *Vicia faba* (Nouzová et al., 1999).

The proposed scenario of satellite DNA evolution from tandem repeats originating in retrotransposon 3′ UTRs is not the only one that could explain sequence similarity between satellites and retrotransposon repeats. There is also the possibility of the reverse process, i.e. the capture of satellite DNA fragments by retrotransposons. The presence of extra open reading frames in the 3′ UTR of Grande, RIRE2 and Retand elements (Martínez-Izquierdo et al., 1997; Ohtsubo et al. 1999; Kejnovský et al., 2006) and of a short stretch of telomeric repeats in the element Pt5 (our observation) can be explained by the uptake of foreign DNA sequences and thus provide support for this explanation. On the other hand, there are strong arguments in favor of the former hypothesis. First, the array of PisTR-A repeats in the pea cosmid c711 represents amplified and homogenized satellite sequence, yet it is surrounded by the retrotransposon 3′ UTR and LTR sequences (Fig. 1). This suggests that the satellite array originated from a shorter one present in the retrotransposon, which is also supported by data for two other investigated repeats. Retroelements containing STR repeat arrays expanded to over 10 kb in length were found in several soybean genomic clones (e.g. AC234042), and a similar observation was made for *P. trichocarpa* elements containing tandem repeat arrays with various degrees of expansion, from a few kb up to 21 kb (AC212921). Second, the sequence diversity of tandem repeats encountered in elements from a single species is more likely to be generated *de novo* during the error-prone process of retroelement replication (Preston, 1996) rather than by capturing separate fragments of many different satellites. For example, there was a large number of different 3′UTR repeats detected in the elements from rice (*O. sativa*) (Table 1), however, no satellites corresponding to these repeats are known from this extensively sequenced genome.

Several hypotheses have been proposed to explain the origin and eventual function of tandem repeats in retrotransposon 3′ UTR regions. These tandem repeats could be involved in the acquisition of new coding information through gene transduction (Sanz-Alferez et al., 2003; Havecker et al., 2004), or facilitate transcription termination through forming stable secondary structures (Martínez-Izquierdo et al., 1997). However, additional coding sequences as well as palindromic tandem repeats have been found only in a small subset of the Tat elements. Alternatively, the repeats could be a consequence of illegitimate recombination or abnormal strand transfers between two element copies co-packaged in a single virus-like particle, which are processes known to generate structural variability of retroelements (Temin, 1993; Sabot et al., 2005; Sabot and Schulman, 2007; Moisy et al., 2008). Although no final explanation of the origin and eventual function of the 3′ UTR repeats in retrotransposon sequences can be provided at the moment, the sequence data gathered in this study should aid in further investigation of this interesting phenomenon.

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